Rate, Factors, and Outcome of Delayed Graft Function After Kidney Transplantation of Deceased Donors

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ABSTRACT

Background. Delayed graft function (DGF) is a frequent complication after kidney transplantation affecting long-term outcome.

Patients and methods. A total of 525 consecutive recipients (age 54.2 ± 13.4 years, 33% female) of kidneys from deceased donors transplanted between 2005 and 2012 were retrospectively examined. DGF was defined as the need of dialysis within the first week after transplantation.

Results. DGF developed in 21.1% (n = 111). Factors associated with DGF ($P \leq .035$, respectively) were recipient body mass index, C-reactive protein of the recipient, residual diuresis, cold ischemia time, donor age, and diuresis in the first hour after transplantation. Median duration of DGF was 16 (2-66) days. Patients after DGF had a significantly lower GFR compared with recipients without DGF either after 3 (32.9 ± 16.5 vs 46.3 ± 18.4 mL/min/1.73 m^2) or after 12 months (38.9 ± 19.3 vs 48.6 ± 20.4 mL/min/1.73 m^2, $P < .001$, resp.). During DGF, 12.4% developed BANFF II and 18.0% BANFF I rejection, 20.2% had signs of transplant glomerulitis (first biopsy), and 16.2% (n = 18) remained on dialysis.

Conclusion. DGF affects 1 out of 5 kidney transplants from deceased donors. Minimizing modifiable risk factors, in particular immunologic risk, may ameliorate the incidence and outcome of DGF. The outcome of DGF depends mainly on the diagnosis of any rejection and worsens upon detection of transplant glomerulitis and pronounced interstitial fibrosis and tubular atrophy (IFTA).

DELAYED graft function (DGF) is a frequent complication after deceased donor kidney transplantation with long-term consequences for the function of kidney transplants including reduced graft survival [1]. DGF occurs in up to 25% of deceased donor kidney transplants [2] but in lower than 5% (3.6% according to US data) of living-donor kidney transplants [3].

Various criteria are used for the diagnosis of DGF. The most commonly used definition describes DGF as the need for dialysis within the first week after transplantation [3,4].

Multiple factors are known to be associated with the presence of DGF, both donor- and recipient-related [2,5]. A well-known risk model for deceased donor renal transplantation was described by Irish et al in 2010 consisting of 11 continuous and 9 categorical variables both from the donor and the recipient [2].

Based on donor and recipient data provided to the Collaborative Transplant Study (CTS) in Germany as well as local laboratory findings, it was the aim of our study to

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determine the frequency and risk factors of DGF over a period of 8 years as well as their outcome within the first year after transplantation.

PATIENTS AND METHODS

In this retrospective study, 525 consecutive recipients (age 54.2 ± 13.4 years, 33% [n = 173] female) of kidneys from 462 deceased donors, transplanted between 2005 and 2012, were examined. DGF was defined by the need for any dialysis treatment within the first week post transplantation and irrespective of the later finding of any rejection. Dialysis was usually necessary in the case of ongoing potassium concentration >6 mmol/L despite conservative treatment, clinical relevant hypervolemia (ie, with dyspnea, hypertension, and growing edema), or strongly increasing retention parameters in the state of permanent oliguria or anuria.

Data was collected at the Kidney Transplant Center of the University Hospital Jena. All organs were allocated by Eurotransplant ensuring that donors were not from prisoners, and were not paid or coerced. The study was conducted according to the 2000 Declaration of Helsinki. The study was approved by the Internal Review Board of ethics (Vote: Number 4221-09/2014). Data collection was based on the questionnaire of the CTS (http://www.ctstransplant.org/) not including the terminal serum creatinine of the donor. All participants gave their informed consent before inclusion. All kidney transplant donors and recipients were of Caucasian descent. Kidney transplants from live donors, double kidney transplants or combined organ transplants were not included. Patients with DGF were followed for 1 year and renal function was evaluated at 3, 6, 9, and 12 months after transplantation. The observation period ended with graft loss, death, or date of the last documented clinical contact with the transplant recipient.

Immunosuppressive Treatment

The immunosuppressive therapy at the Kidney Transplant Center Jena is carried out according to a standardized scheme consisting of corticosteroids, mycophenolate mofetil (MMF), and a calcineurin inhibitor. The scheme divides into low and high immunologic risk, as well as patients ≥65 years who are transplanted within the Eurotransplant Senior Programme receiving organs from donors aged ≥65 years. All patients receive preoperatively 500 mg of methylprednisolone intravenously as induction therapy. High-risk patients (any retransplantation, panel reactive antibodies >5%, and a sum of >2 HLA-mismatches in the loci B and DR or 2 in DR) are induced additively with antithymocyte globulin (ATG, Thymoglobulin, Genzyme Europe B.V., Amsterdam, Netherlands) over 3 days starting intraoperatively at a cumulative dosage of 3.5 mg/kg bodyweight and receive tacrolimus, MMF, and steroids as maintenance immunosuppression. Patients in the low-risk stratum are treated with cyclosporine, MMF, and steroids in the long-term as also patients within the Eurotransplant Senior Programme. Seniors additionally receive basiliximab for 2 times (day 0 and day 4, 20 mg each) as induction therapy.

Before the diagnosis of DGF, any exclusion of surgical causes (ie, hematoma, urinary retention, ischemia) of transplant dysfunction

Table 1. Baseline Data of 525 Transplant Recipients and 462 Donors

Variable	Total n = 525	DGF n = 111	No DGF n = 414	*P* Value
Recipient characteristics				
Sex, female (%)	173 (33.0)	39 (35.1)	134 (32.4)	.582
Age, y	56 (14-82)	55 (17-82)	56 (14-77)	.845
BMI, kg/m^2	25.8 (15.3-44.5)	27.9 (16.7-41.8)	25.3 (15.3-44.5)	**<.001**
Duration of dialysis, mo	65 (4-281)	72 (12-281)	63 (4-226)	**.027**
Residual diuresis, mL/d	200 (0-3000)	0 (0-2500)	300 (0-3000)	**<.001**
Diuresis first hour post transplant, mL	70 (0-1900)	0 (0-1300)	100 (0-1900)	**<.001**
Diuresis first day post transplant, mL/h	62.8 (0-843)	14.6 (0-314)	77.1 (0-843)	**<.001**
eGFR preoperative, mL/min/1.73 m^2	6.8 (2.6-11.6)	6.4 (3.3-19.3)	6.9 (2.6-11.6)	**.027**
Cold ischemia time, h	10.4 (1.5-30.5)	11.2 (1.5-30.3)	10.2 (4.5-30.5)	**.039**
Warm ischemia time, h	37 (17-77)	40 (24-73)	37 (17-77)	**<.001**
CRP preoperative, mg/L	4.2 (2-76.8)	5.9 (2-76.8)	3.7 (2-63)	**<.001**
CRP day 1 post transplant, mg/L	27 (2-250)	34 (2-250)	26 (2-229)	.221
Leukocytes day 1 post transplant, Gpt/L	12.3 (3.3-29.2)	13.5 (5.2-29.2)	12 (3.3-28.1)	**.005**
HLA mismatches				.467
<3 (%)	189 (36.0)	42 (37.8)	147 (35.5)	
3-4 (%)	238 (45.3)	40 (36.0)	198 (47.8)	
>4 (%)	98 (18.7)	29 (26.1)	69 (16.7)	
Immunosuppression				.735
Low risk (%)	219 (41.9)	45 (40.5)	174 (42.2)	
High risk (%)	154 (29.4)	33 (29.7)	121 (29.4)	
Senior* **(%)**	150 (28.7)	33 (29.7)	117 (28.4)	
Missing (%)	2 (0.4)	0	2 (0.5)	
Donor characteristics				
Sex, female (%)	173 (33.0)	39 (35.1)	134 (32.4)	.582
Age, y	56 (14-82)	55 (17-82)	56 (14-77)	.845
BMI, kg/m^2	25.7 (17.1-58.8)	26.1 (18.3-57.8)	25.5 (17.1-58.8)	**.036**

Bold formatting shows significant *P* values ($P < 0.05$).
Values given as median (min-max) or number (%), as appropriate.
Abbreviations: BMI, body mass index; CRP, c-reactive protein; DGF, delayed graft function; eGFR, estimated glomerular filtration rate; HLA, human leucocyte antigen.
*Patients transplanted within the Eurotransplant Senior Programme.

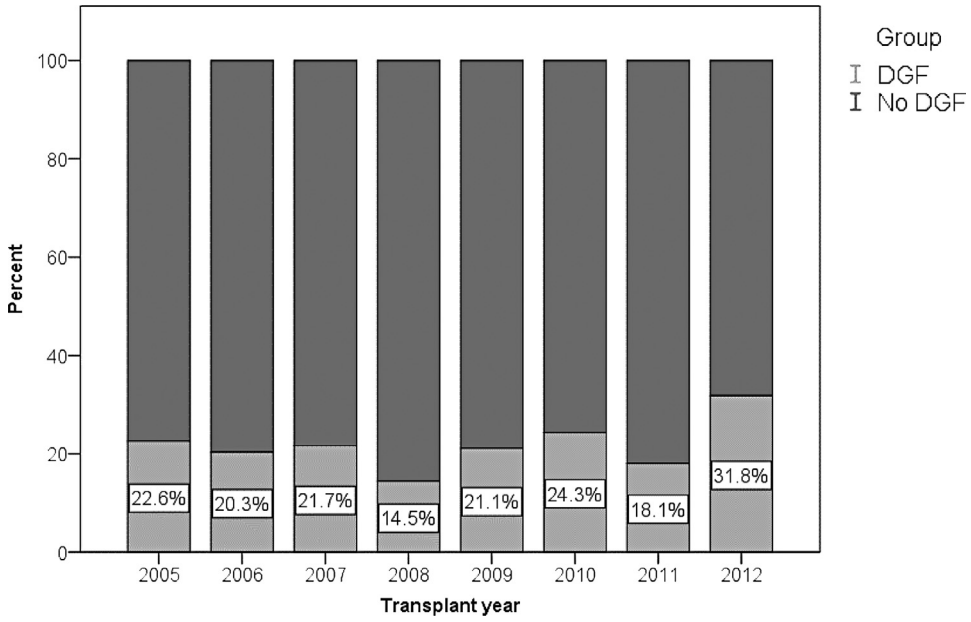


Fig 1. DGF rate per year over 8 years in 525 consecutive recipients of kidney transplants. DGF, delayed graft function.

was obligatory performed. Biopsy proven rejections were classified according to the Banff classification of 2007 [6].

The day of recovery of the organ was set as the second day after the last documented dialysis session.

STATISTICAL METHODS

The statistic program SPSS 23.0 (IBM Corporation) was used. The level of significance was set at 5 percent ($P < .05$). In order to find significant differences between 2 groups, the χ^2 test was used for categorical variables and the Mann-Whitney U test for continuous parameters. For more than 2 groups, the Kruskal-Wallis test was used. Using univariate and stepwise multivariate logistic regression analysis (forward and backward selection), factors for the occurrence and outcome of DGF including permanent nonfunction (PNF) were examined. In the latter, histologic findings of the first kidney biopsy were included. Parameters with a P value of <.1 (for the occurrence of DGF) and <.15 (for the occurrence of PNF) in univariate analysis were included in the initial multivariate models. Gain of function of transplanted kidneys with DGF is shown as Kaplan-Meier curve.

RESULTS

Baseline Data and Rate of DGF

Baseline data and selected donor characteristics in patients with and without DGF are given in Table 1. Overall, DGF developed in 21.1% (n = 111) of all renal transplants (n = 525) but with fluctuations over the years (Fig 1). The mean donor age was 54.6 ± 14.4 years, the mean recipient age 54.2 ± 13.4 years. The recipient age increased significantly from 2005 until 2012 ($P = .004$), whereas the donor age remained stable ($P = .053$). Recipients with DGF had a significantly higher BMI, received organs from obese donors, had a longer preoperative dialysis time, a lower residual diuresis, longer cold and warm ischemia times, as well as preoperatively higher values of C-reactive protein (CRP) ($P \leq .039$, resp., Table 1).

Factors for the Occurrence of DGF

Factors significantly associated with the onset of DGF ($P \leq .035$) are shown in Table 2. Although significant in univariate analysis, an effect of donor BMI (odds ratio [OR] 1.048 per increase of 1 kg, 95% confidence interval [CI]

Table 2. Multivariate Logistic Regression Model for the Presence of DGF, n = 111 (21.1%) of 525 Transplant Recipients (Final Model)

Variable	RC	SE	OR	95% CI	*P* Value
Recipient BMI (kg/m^2)	0.067	0.027	1.070	1.015-1.128	**.012**
Donor BMI (kg/m^2)	0.043	0.026	1.044	0.993-1.098	.090
Cold ischemia time (h)	0.055	0.026	1.057	1.004-1.112	**.035**
Residual diuresis (per increase of 100 mL/d)	−0.083	0.024	0.921	0.879-0.965	**.001**
Recipient age (y)	−0.022	0.012	0.979	0.956-1.002	.071
Donor age (y)	0.026	0.012	1.026	1.002-1.051	**.035**
Recipient CRP (preoperative) (per 1 mg/L)	0.026	0.012	1.026	1.002-1.051	**.035**
Diuresis in the first hour (per increase of 100 mL)	-0.492	0.168	0.611	0.440-0.850	**.003**

Adjusted for all of the above variables as well as duration of dialysis and warm ischemia time
Abbreviations: BMI, body mass index; CI, confidence interval; CRP,C-reactive protein; OR, odds ratio; RC, regression coefficient; SE, standard error.

Table 3. Histological Findings of the First Biopsy During DGF (n = 111) in Patients With and Without Later PNF

Variable	Total n = 111	PNF n = 18	No PNF n = 93	*P* Value
Acute tubular damage	74 (83.1)	14 (82.4)	60 (83.3)	1.000
BANFF I rejection	16 (18.0)	0	16 (22.2)	**.035**
BANFF II rejection	11 (12.4)	3 (17.6)	8 (11.1)	.433
Transplant glomerulitis	18 (20.2)	6 (35.3)	12 (16.7)	.101
Relevant signs of IFTA*	35 (39.3)	11 (64.7)	24 (33.3)	**.026**
Glomerulosclerosis	30 (33.7)	9 (52.9)	21 (29.2)	.087
Nephrosclerosis	36 (40.4)	10 (58.8)	26 (36.1)	.104
Thrombotic microangiopathy	1 (1.1)	1 (5.9)	0	.191
Calcineurin-inhibitor toxicity	0	0	0	
Viral infection	0	0	0	

Values given as number (%).
Abbreviations: IFTA, interstitial fibrosis and tubular atrophy; PNF, permanent nonfunction.
*Only if rated at least as moderate, or comparable to ≥25%, as given in the medical report.

1.007-1.09, $P = .023$), dialysis duration (OR 1.007 per increase of 1 month, 95% CI 1.001-1.013, $P = .015$), and warm ischemia time (OR 1.040 per increase of 1 minute, 95% CI 1.018-1.063, $P < .001$) could not be confirmed in the multivariate analysis.

Biopsy Results

Eighty percent (n = 89) of all DGF patients received at least 1 kidney biopsy during the postoperative course. An initial biopsy was performed after 9.5 ± 2.7 days. Patients with PNF received biopsies in 94.4% and patients without PNF in 77.4%. Nearly 58% (n = 64) of the DGF group received a second biopsy and 37% a third biopsy. Patients with later PNF (n = 18) received a second biopsy in 83.3% (n = 15), whereas this was the case in only 52.7% (n = 49) of those without PNF (n = 93, $P = .019$). Histologic findings of the first biopsy are shown in Table 3.

Resolution of DGF

Achievement of transplant function was observed in 93 of 111 transplants with DGF (83.8%). The median time to functional recovery was 16 (2-66) days. Ninety-six percent of all patients with recovering kidney function achieved function after 33 days (Fig 2).

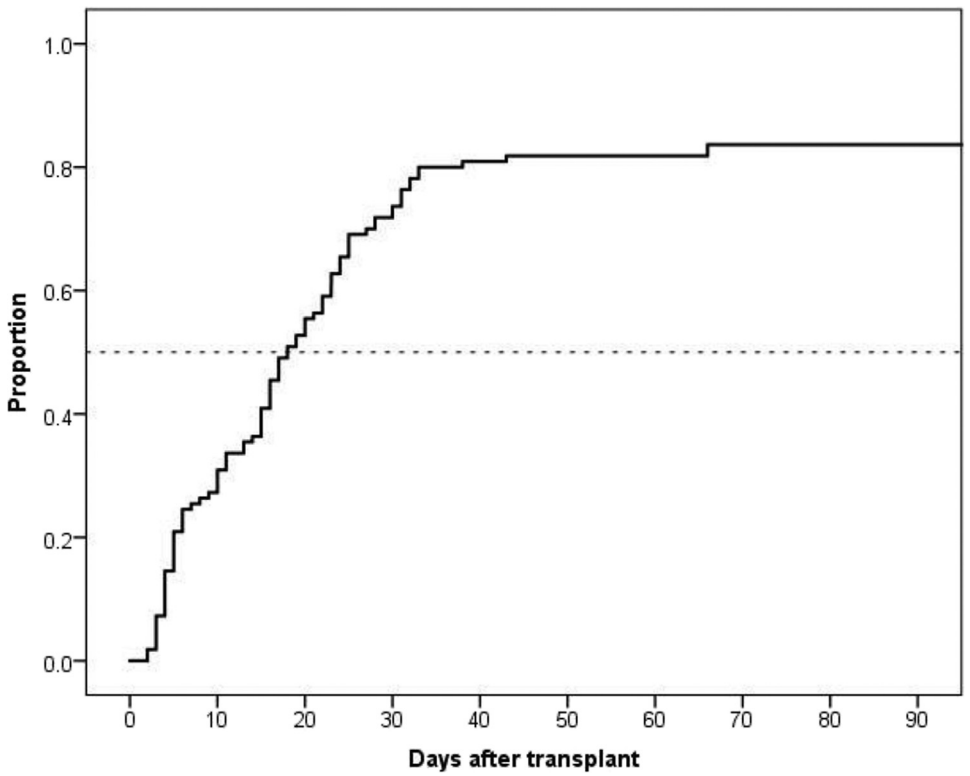


Fig 2. Achieving of renal graft function over time in 111 recipients with DGF, n = 18 (16.2%) remained on dialysis.

Table 4. Final Model of Multivariate Logistic Regression Analysis on the Occurrence of PNF (n = 18) in 121 Recipients With DGF

Variable	RC	SE	OR	95% CI	*P* Value
Immunosuppressive regimen					
Low-risk (reference)					.082
High risk (vs low risk)	1.841	0.864	6.301	1.159-34.257	**.033**
Senior* (vs low risk)	0.530	0.932	1.699	0.273-10.563	.570
Transplant glomerulitis (yes vs no)	1.645	0.819	5.180	1.040-25.804	**.045**
Interstitial fibrosis/tubular atrophy† (yes vs no)	2.308	0.805	10.055	2.076-48.716	**.004**
Leukocytes at day 1 (per increase of 1 Gpt/L)	−0.193	0.094	0.824	0.685-0.992	**.040**

Adjusted for all of the above parameters as well as BMI grouping of the donor, nephrosclerosis, presence of glomerulosclerosis, PRA (highest value).
Abbreviations: CI, confidence interval; OR, odds ratio; RC, regression coefficient; SE, standard error.
*Patients transplanted within the Eurotransplant Senior Programme.
†Only if rated at least as moderate, or comparable to ≥25%, as given in the medical report.

PNF

Transplants that had no function even after 90 days were referred to as PNF. A total of 18 transplant recipients (16.2%) with DGF remained on dialysis (Fig 2). Significant factors for the presence of PNF ($P \leq .045$) are depicted in Table 4. In contrast to transplant glomerulitis ($P = .093$), the finding of interstitial (BANFF I, $P = .998$) or vascular rejection (BANFF II, $P = .465$) did not reach the level of statistical significance in univariate modeling ($P < .15$) to be included in the initial multivariate model.

Follow-up of DGF Patients in the First Year After Transplantation

During follow-up, recipients with DGF and functioning graft showed a significantly lower estimated glomerular filtration rate compared with recipients without DGF after 3, 6, 9, and 12 months ($P \leq .001$, resp., Fig 3). Within the first year after transplantation, 42 (45.2%) from the 93 DGF patients with functioning graft developed acute biopsy-proven rejections, 1 graft was lost (1.1%), and 7 patients (7.5%) died. The causes of death were sepsis (n = 4), heart failure (n = 1), pulmonary embolism (n = 1) and one was unknown. Fifteen grafts of patients with PNF (n = 18) were surgically removed during the first year after transplantation.

DISCUSSION

DGF is a common complication after kidney transplantation and a prognostic factor [4,7–9]. It is associated with higher transplant costs [7] and leads to an increased

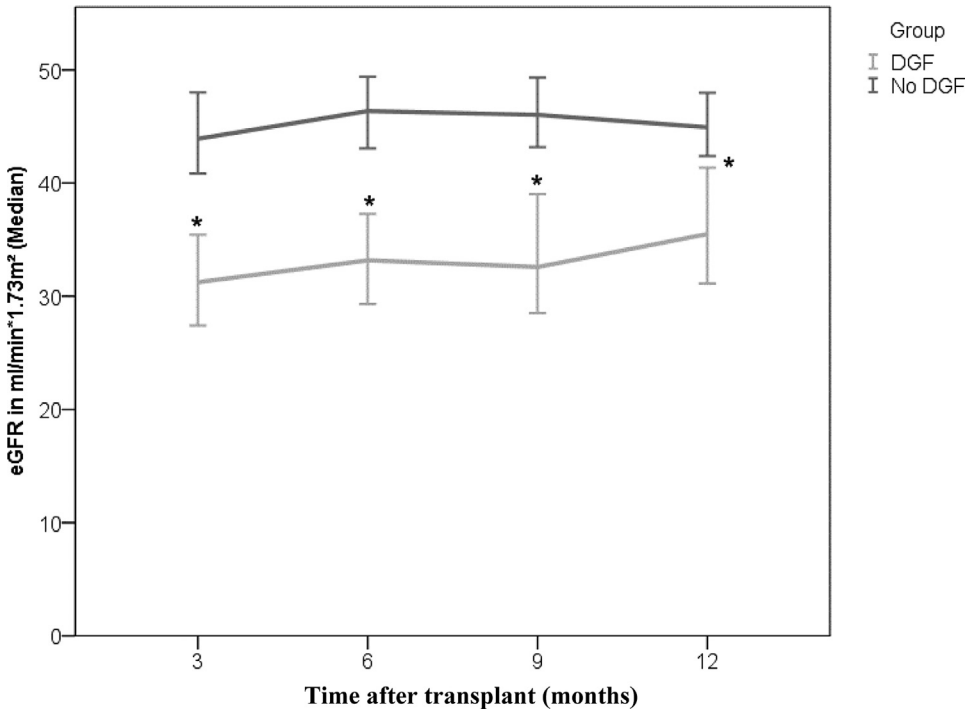


Fig 3. eGFR at 3, 6, 9, and 12 months after transplantation in patients with and without DGF (*$P < .05$, error bars: 95% CI). DGF, delayed graft function; eGFR, estimated glomerular filtration rate.

risk of rejection, impaired organ function, and premature graft lost [7,9,10]. The rate of DGF is 2% to 50% of all transplants [4,8,11,12]. Extended or expanded criteria donors and donations after cardiac death are particularly affected [5,13]. The rate of DGF in living-donor kidney transplantation was 3.6% [3].

Using the common definition of DGF as the need for dialysis within the first week post transplantation [2,11], the DGF incidence of 21.1% recorded here was slightly lower than the rate of 25.7% reported by Irish et al between 2003 and 2006 in the United States [2]. A Dutch study found a markedly higher DGF prevalence of 37% between 2002 and 2012, although patients with PNF were excluded [12]. Instead, our study encompasses 18 patients with ultimately PNF. Although our study did not include donors after cardiac arrest it included 150 patients transplanted within the Eurotransplant Senior Programme [14] who are classified as expanded criteria donors due to some definition [15]. Nevertheless, the rate of DGF was nearly the same as recently published by Lim et al who found a rate of 19.5% in Australia and New Zealand [10]. The median duration of DGF in this study was shorter (7 days) as compared with 16 days in the current study.

Despite the usefulness [2,5,16] and reproducibility [12] of existing models for the prediction of DGF, their clinical use is limited due to the prevailing lack of organs against the background of mostly not ideal recipients. Moreover, predictors for DGF are mostly performant in the population in which they were derived but less so in external validations [13]. Two monocentric Belgian studies [13,17] found the model of Irish et al [2,11] as the most consistent with an acceptable discriminative capacity, but the prediction for expanded criteria donors was poor [13]. The model comprises cold ischemia time (CIT), donor terminal serum creatinine, recipient body mass index (BMI), donor age, and organs after cardiac arrest of the donor [2]. Based on this model, Decruyenaere et al identified donor age, donor BMI, terminal serum creatinine of the donor, donation after cardiac death as well as recipient BMI, duration of dialysis, preoperative diastolic blood pressure, and a reduced cardiac function of the recipient as significant factors for DGF [17]. Both donor and recipient BMI also appeared in the final model of the present study, although the donor BMI failed statistical significance.

Although our study did not evaluate terminal serum creatinine of the donor and did not include any transplantation after cardiac arrest, the factors CIT, recipient BMI, and donor age also could be identified. In a French study, Chapal et al developed a simple DGF score of 5 variables containing donor age, serum creatinine of the donor, CIT, the BMI of the recipient, as well as the nonuse of antithymocyte globulin [18]. Lafranca et al investigated the influence of low BMI recipients (<30 kg/m^2) by meta-analysis and were able to confirm a benefit with regard to the occurrence of DGF [19]. Higher recipient age also was described as an independent risk factor for the occurrence of DGF [20].

In the present work, growing residual diuresis of the recipient before transplantation could be identified as a protective factor for DGF. Since residual diuresis is closely associated with residual renal function and thus the duration of dialysis, it is quite in line with previous findings who identified prior dialysis time as an associate with DGF [17]. Dialysis time could not be confirmed as an independent risk factor in our model, possibly because of its association with residual diuresis. Nevertheless, residual diuresis might be an underestimated and more sensitive factor belonging the individual consequences of longer dialysis time (ie, vascular disease, inflammation, and volume fluctuations during hemodialysis) contributing to transplant-associated ischemia-reperfusion damage and thus influencing DGF [21]. In anuric dialysis patients, increased inflammation and oxidative stress was observed [22]. The possibility of DGF was doubled if residual diuresis was <500 mL per day [23].

Active inflammation of the recipient is known to be a relative contraindication to renal transplantation [24]. So far, there is no clear evidence if inflammation could really aggravate DGF risk. Reek et al found a significant increase in CRP during the presence of DGF, but it was not independently associated with DGF [25]. Instead, CRP (determined right before renal transplantation) was an independent factor for the development of DGF in our study. In contrast, Lauzurica et al found no association between preoperative CRP and DGF, but an association with the increased cytokine TNF-alpha possibly outweighting CRP [26]. This might also hint to a possible impact of preoperative inflammation for the development of DGF. In animal model, post-transplantation vascular changes could be linked to occult pretransplantation systemic inflammation [27]. Furthermore, elevated CRP levels predicted independently a more accelerated deterioration of graft function [28].

Among the postoperatively raised markers, diuresis within the first hour after transplantation indicated relatively safe against DGF confirming previous findings that demonstrated a predictive value of intraoperative diuresis for the onset of DGF [29]. The significance of early intra- and postoperative diuresis is studied relatively little concerning DGF, as the onset of diuresis speaks rather against DGF. However, early diuresis also could be driven by residual diuresis, especially with perioperative volumetric stimulation. Nevertheless, although this factor represents a postoperative determined sign precluding any preoperative decision making, it enables an early postoperative evaluation for the probability of a DGF as our data suggest.

DGF is known to cause detrimental influence on the long-term graft function and graft half-life [4,5,9] as the present data confirm for the first year after transplantation. The cause may be a protracted and sometimes irreversible damage to the graft as reflected by a higher degree of interstitial fibrosis and tubular atrophy (IFTA) as the biopsy

data prove, especially for patients with PNF. In this regard, the duration of DGF with a median recovery time of 16 days is likely to have played a role, as DGF of more than 6 days decreases long-term transplant survival [7]. The comparably longer duration of DGF also may have increased the risk of transplant glomerulitis and rejection since the duration of DGF could be linked with the risk of rejection and graft survival [10].

Based on our data, it was relatively unlikely that an organ with DGF would start its function after more than 33 days post transplant. PNF had 16.2% (n = 18) of all patients with DGF, which was 3.4% of all grafts examined. Due to older data from the 1990s the rate of PNF was up to 14% of all transplants [30]. As independent risk factors, a high panel-reactive antibody value (>50%) of the recipient, higher mismatch number, retransplantation, and cyclosporine-free immunosuppression were identified at that time [30]. In our study, belonging to the high-risk immunosuppressive regimen group (indicating higher immunologic risk, as described above), lower leukocyte counts on day 1 post-operatively, and the histologic evidence of graft glomerulitis and pronounced IFTA in the first biopsy were significant factors for the presence of PNF. Another retrospective study also found a higher DGF rate in recipients with histologic evidence of transplant glomerulitis [31]. Moreover, transplant glomerulitis is known to be strongly associated with increased risk of graft failure [32] as confirmed by the early post-transplant data of the current study. Obviously, the stronger immunosuppressive therapy of our immunologic high-risk patients including ATG could only partially compensate for that risk contrary to other data who described ATG to be protective against DGF [18]. Thus, the individual immunologic risk stratification of transplant recipients remains a persistent challenge in preventing and managing the risk for DGF and associated PNF, as also reflected by the follow-up data of the first year showing a high rate of rejection.

Our biopsy data in DGF confirm a rejection rate of 30% already in the first biopsy underlining the urgent need for (repetitive) transplant biopsies during DGF. Since the presence of rejection was not predictive for PNF, rejections were apparently treatable, in particular T-cell mediated BANFF I rejections [6]. After all, up to 50% of DGF patients were affected by BANFF I or II rejection or transplant glomerulitis highlighting the immunologic cause and risk of DGF (ie, by antigen exposure due to graft damage) [9,33]. Quantification of IFTA is another benefit of graft biopsy because the probability of PNF was 10 times more likely if DGF was associated with the histologic finding of pronounced IFTA confirming previous findings [34,35]. Unexpectedly, higher leukocyte counts measured on the first postoperative day were a protective factor for PNF. The most probable explanation of this association may be found in the administration of ATG in patients with high-risk immunosuppression. ATG is known to cause leukopenia and other cytopenias [36] that are often to be seen at the days of administration.

DGF is described as a major risk factor for the occurrence of later acute rejection [10,33,37]. In the present study, DGF patients with already functioning graft had a high rejection rate of 45% within the first year after transplantation. Quiroga et al found a rejection rate of 54% in DGF patients [38]. Wu et al were able to demonstrate an increase in the risk for acute rejection after DGF up to 4 years after transplantation [33]. Thereby, a close monitoring of patients after DGF is recommended.

Limitations of the present study are due to a lack of standardization of the definition of delayed graft function. The definition used by the current study comprises the fact that the indication for dialysis is a highly individual and, in part, center-specific decision [39]. As already mentioned, the final serum creatinine of the donor could not be recorded due to a certain lack of data. Nevertheless, the study has the strength that it provides valuable insights into the rate, course, and outcome of DGF considering biopsy findings and patients with PNF over 8 years.

CONCLUSION

The presence of DGF is the result of multiple factors and affects 1 in 5 kidney transplants of deceased kidney donors. DGF is associated with a poorer graft function. Appropriate donor and recipient selection and matching in the process of organ acceptance taking into account immunologic risk, recipient BMI, CIT, donor age, as well as residual diuresis and CRP of the recipient could help to reduce the incidence of DGF and thus improve long-term kidney function. The outcome of DGF depends mainly on the exclusion of any rejection and worsens upon detection of transplant glomerulitis and pronounced IFTA. Diuresis within the first hour can be valued as an early sign against DGF.

DATA AVAILABILITY

Data will be made available on request.

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